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- **Demonstrative Example 5 (Fictitious):** The A/E could be asked to design an automatic fire alarm and suppression system for a historic building that protects the building and its occupants from fire while maintaining the building's characteristic features and historic fabric while following the Secretary of the Interior's Standards and Guidelines for Archeology and Historic Preservation.

I. DISCIPLINES

- **Required Engineering Disciplines.** Required engineering disciplines include, but are not limited to, registered professional engineers who possess professional licenses in at least one state or protectorate and who are experienced at practicing each specific required engineering discipline. Specific engineering disciplines are listed below and described in Attachment A in the statement of work (SOW): Principal Engineer (Senior Engineer)
 - Principal Engineer (Senior Engineer)
 - Senior Consultant (Principal Project Manager)
 - Senior Project Manager
 - Electrical Engineer
 - Electrical Designer
 - Mechanical Engineer
 - Mechanical Designer
 - Civil Engineer
 - Structural Engineer
 - Historic Architect
 - Fire Protection Engineer 1
 - Fire Protection Engineer 2
 - Fire Protection Engineer in Training (EIT)
 - Fire Protection Designer
 - Senior CAD Technician
 - CAD Technician
 - Computer Network Administrator
 - Cost Estimator
- A. Incidental Services: Incidental Services, e.g., AutoCAD drafting, other architectural and engineering disciplines, and 3D modeling, GIS, Microsoft Office applications, Adobe Acrobat, copying/printing, material support for field-assigned personnel may be required in the performance of specific task order requirements. Engineering or technical support documents that address the restoration of surfaces (plaster/drywall repair, masonry, carpentry, backfill, etc.) may be required but actual facilities related construction work is not within this scope.
- B. Other disciplines not listed but required for completion of a specific task associated with a specific project will be added as necessary.

II. PRIMARY SERVICES REQUIRED

- A. For the purposes of this contract, the services to be provided will primarily consist of the following:
 1. Inventory of fire protection systems and certain attributes;
 2. Fire protection engineering and conceptual design services;
 3. Construction oversight, system acceptance, and collaboration for the issuance of fire and life safety certificates of compliance;
 4. Life Safety and Fire Risk Assessments;
 5. Cost Estimating; and
 6. Construction Field Engineering (i.e. construction engineering, inspection, on-site testing, safety

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- Task orders issued under this item are primarily for the purpose of providing construction phase oversight for the FCO/AHJ of contract installer’s work, to commission fire protection systems, and to recommend the issuance of a fire and life safety certificate of compliance to the FCO/AHJ.

3. Life Safety and Fire Risk Assessments;

- Task orders issued under this item are primarily for the production of a Life Safety and Fire Risk Assessment per DO/RM 58: Structural Fire Management, the production of a physical survey, cost estimates, and accompanying report, and the provision of a complete evaluation of each park structure’s construction, function, operational support systems, and occupancy as they impact fire protection and life safety. The report shall identify risks to life, property, or park mission from the effects of potential fire incidents. The report shall serve as an NPS management reference document for planning and prioritizing projects for short-term work to maintain satisfactory facility safety, and to plan for long-range decisions regarding renovation, reinvestment, and preservation. Specific language regarding the assessment of collections and/or historic objects may be included in task orders.

2.0 Disciplines

1. Required Disciplines: Required engineering disciplines include, but are not limited to, registered professional engineers who possess professional licenses in at least one state or protectorate and who are experienced at practicing each specific required engineering discipline. Specific disciplines required are listed below and described in Attachment A:

- Principal Engineer (Senior Engineer)
- Senior Consultant (Principal Project Manager)
- Senior Project Manager
- Electrical Engineer
- Electrical Designer
- Mechanical Engineer
- Mechanical Designer
- Civil Engineer
- Structural Engineer
- Historic Architect
- Fire Protection Engineer 1
- Fire Protection Engineer 2
- Fire Protection Engineer in Training (EIT)
- Fire Protection Designer
- Senior CAD Technician
- CAD Technician
- Computer Network Administrator
- Cost Estimator

2. Incidental Services: Incidental Services, e.g., AutoCAD drafting, other architectural and engineering disciplines, and 3D modeling, GIS, Microsoft Office applications, Adobe Acrobat, copying/printing, material support for field-assigned personnel may be required in the performance of specific task order requirements. Engineering or technical support documents that address the restoration of surfaces (plaster/drywall repair, masonry, carpentry, backfill, etc.) may be required but actual facilities related construction work is not within this scope.

3.0 Professional Licenses and Registrations for “Required Disciplines”

At a minimum, the contractor must be able to provide fire protection engineers with experience in condition assessments, work in government buildings, and work involving historic buildings, collections, and archives. Assessments will require a thorough knowledge of the most recent edition of the International Code Council (ICC) codes to include the International Fire Code (IFC) and the International Wildland-Urban Interface Code (IWUIC), and the National Fire Protection Association (NFPA) codes and standards referenced therein, including NFPA 70: National Electrical Code (NEC). Specific policy related requirements that differ from code, such as those in DO/RM 58: Structural Fire Management or the Structural Fire Design Guide, will be provided when needed. All work where a fire protection engineer is required is to be accomplished by a registered professional fire protection engineer in the United States. All task orders will require the contractor to appoint a senior fire protection engineer as the project manager who will be responsible for review of all submittals prior to submission to the NPS. Particular tasks may require specific skills and certifications, which will be listed in the task orders when issued.

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Attachment A

Descriptions of Required Disciplines

1) Principal Engineer (Senior Engineer)

General Experience: Principal engineers (senior engineers) have 10 plus years of practical experience.

Functional Responsibility: Responsible for playing a lead role in designing fire protection and life safety systems. Serves as a senior technical resource for project teams, providing expert guidance and addressing complex engineering challenges. Functions as the engineer of record and may supervise multiple project teams.

Minimum Education: Bachelor's degree

Minimum Licensing: Licensed Professional Engineer (Fire Protection Engineering Exam) in any state or territory, prefer licensed in state where work is being performed.

2) Senior Consultant (Principal Project Manager)

General Experience: Senior consultants (principal project managers) have 10 years of practical experience.

Functional Responsibility: Responsible for overseeing complex projects from conception through completion. Develops comprehensive project plans, timelines, and milestones; allocates resources; manages project budgets; monitors progress; identifies risk and maintains stakeholder engagement. Oversees and mentors junior project managers.

Minimum Education: Associate degree or equivalent experience

3) Senior Project Manager

General Experience: Senior project managers have five to seven years of practical experience.

Functional Responsibility: Responsible for overseeing complex projects from conception through completion. Develops comprehensive project plans, timelines, and milestones; allocates resources; manages project budgets; monitors progress; identifies risk and maintains stakeholder engagement. Oversees and mentors junior project managers.

Minimum Education: Associate degree or equivalent experience

4) Electrical Engineer

General Experience: Electrical engineers have at least four years of practical experience after becoming a licensed Professional Engineer.

Functional Responsibility: Responsible for designing fire protection and life safety systems in accordance with fire codes and standards. Works closely with architects, builders, and design professionals. Conducts fire safety evaluations of systems and equipment to identify potential hazards and recommend improvements.

Minimum Education: Bachelor's degree

Minimum Licensing: Licensed Professional Engineer (Fire Protection Engineering Exam) in any state or territory, prefer licensed in state where work is being performed.

5) Electrical Designer

General Experience: Electrical designers have three to five years of practical experience after certification.

Functional Responsibility: Responsible for creating detailed blueprints and schematics for electrical systems using CAD software to produce accurate designs to meet project specifications in alignment with associated fire codes and standards. Oversees the electrical design process from conception to completion, managing timelines and budgets while ensuring projects meet quality standards.

Minimum Education: Bachelor's degree

6) Mechanical Engineer

General Experience: Mechanical engineers have at least four years of practical experience after becoming a licensed Professional Engineer.

Functional Responsibility: Responsible for designing and developing mechanical systems and devices, including fire protection and life safety systems. Collaborates with other engineering professionals to ensure mechanical components are integrated into larger building systems. Performs analytical and computational assessments to meet functional requirements, safety standards, and fire codes and standards.

Minimum Education: Bachelor's degree

Minimum Licensing: Licensed Professional Engineer (Fire Protection Engineering Exam) in any state or territory, prefer licensed in state where work is being performed.

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7) Mechanical Designer

General Experience: Mechanical designers have two to five years of practical experience after certification.

Functional Responsibility: Responsible for creating and developing mechanical systems and products, utilizing engineering principles and CAD software. Creates detailed design drawings and specifications for mechanical equipment, tools, and systems, including fire protection and life safety systems. Prepares technical reports and manuals to support the manufacturing and assembly process.

Minimum Education: Associate's degree

8) Civil Engineer

General Experience: Civil engineers have at least four years of practical experience after becoming a licensed Professional Engineer.

Functional Responsibility: Responsible for designing, building, supervising, operating, and maintaining large construction projects and systems. Manages the repair and replacement of private and public infrastructure. Ensures building designs and systems meet fire safety requirements that are in compliance with fire codes and standards.

Minimum Education: Bachelor's degree

Minimum Licensing: Licensed Professional Engineer (Fire Protection Engineering Exam) in any state or territory, prefer licensed in state where work is being performed.

9) Structural Engineer

General Experience: Structural engineers have four years of practical experience after becoming a licensed Professional Engineer.

Functional Responsibility: Responsible for ensuring compliance with prescriptive code compliance for fire resistance ratings of building elements. Creates detailed designs for buildings and other structures and analyzes the strength and stability of these designs using engineering principles and software tools. This can include calculating loads, stresses, and material strengths to ensure structures can withstand environmental forces and human use.

Minimum Education: Bachelor's degree

Minimum Licensing: Licensed Professional Engineer (Fire Protection Engineering Exam) in any state or territory, prefer licensed in state where work is being performed.

10) Historic Architect

General Experience: Historic architects have at least five years of experience as an architect and at least one year of experience with preservation projects.

Functional Responsibility: Responsible for conserving, restoring, and rehabilitating historic buildings by assessing landmark structures, developing restoration plans, and overseeing construction to meet preservation standards, while balancing life safety and sustainability. Researches the historical and architectural value of buildings. Coordinates with construction teams to ensure the use of appropriate techniques and materials that respect the historical integrity of structures.

Minimum Education: Bachelor's degree in architecture

Minimum Licensing: Licensed Architect in any state or territory, prefer licensed in state where work is being performed.

11) Fire Protection Engineer 1

General Experience: A fire protection engineer 1 has 3 or more years of practical experience after licensed Professional Engineer.

Functional Responsibility: Responsible for applying fire protection engineering principles and techniques in the design, development, integration, analysis, operation, and evaluation of fire protection and life safety without supervision. May supervise small project teams.

Minimum Education: Bachelor's degree

Minimum Licensing: Licensed Professional Engineer (Fire Protection Engineering Exam) in any state or territory, prefer licensed in state where work is being performed.

12) Fire Protection Engineer 2

General Experience: A fire protection engineer 2 has 0 to 3 years of practical experience after becoming a licensed Professional Engineer.

Functional Responsibility: Responsible for applying fire protection engineering principles and techniques in the design, development, integration, analysis, operation, and evaluation of fire protection and life safety with little supervision. Prepares design materials for basic design assignments, including basic analysis, equipment research, and design calculations.

Minimum Education: Bachelor's degree

Minimum Licensing: Licensed Professional Engineer (Fire Protection Engineering Exam) in any state or territory, prefer licensed in state where work is being performed.

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13) Fire Protection Engineer in Training (EIT)

General Experience: A fire protection engineer 3 has less than four years of practical experience as an EIT. Works under the direct supervision of a Licensed Professional Engineer in Fire Protection Engineering.

Functional Responsibility: Responsible for using knowledge of fire protection engineering principles and techniques in the design, development, integration, analysis, operation, and evaluation of fire protection and life safety with supervision. May report to fire protection engineers as a project manager.

Minimum Education: Bachelor's degree

Minimum Licensing: No licensing required. Should be an Engineer in Training (EIT) working toward becoming a licensed Professional Engineer.

14) Fire Protection Designer

General Experience: Fire protection designers are fire protection engineers with two to five years of experience or NICET (National Institute for Certification in Engineering Technologies) Level III (or higher) Technicians for the type of system being designed.

Functional Responsibility: Responsible for the design of fire protection systems in buildings and other structures, including fire alarms, suppression systems, and other fire protection equipment to ensure the safety of buildings and occupants. Creates detailed diagrams and blueprints of fire protection systems using CAD software.

Minimum Education: Bachelor's degree

15) Senior CAD Technician

General Experience: Senior CAD technicians have over five years of practical experience after certification.

Functional Responsibility: Responsible for creating high-quality, accurate, and detailed technical drawings, plans, and schematic designs for fire protection and life safety systems, manufacturing, or construction projects compliant with industry standards and code requirements. Collaborates with architects, engineers, designers, project managers, and other members of a project team to understand a project's design requirements. Oversees and/or mentors junior CAD technicians.

Minimum Education: Associate degree or equivalent experience

16) CAD Technician

General Experience: CAD technicians have three to five years of practical experience after certification.

Functional Responsibility: Responsible for creating high-quality, accurate, and detailed technical drawings, plans, and schematic designs for fire protection and life safety systems, manufacturing, or construction projects compliant with industry standards and code requirements. Collaborates with architects, engineers, designers, project managers, and other members of a project team to understand project design requirements.

Minimum Education: Associate degree or equivalent experience

17) Computer Network Administrator

General Experience: Computer network administrators have three to five years of practical experience after certification.

Functional Responsibility: Responsible for managing and maintaining computer networks, ensuring network infrastructure operates smoothly and efficiently, maintaining cybersecurity, and monitoring and troubleshooting network performance.

Minimum Education: Bachelor's degree in computer science, information technology, or a related field or practical experience

18) Cost Estimator

General Experience: Cost estimators have five years or more of practical experience after certification.

Functional Responsibility: Responsible for estimating the cost of designing fire protection and life safety systems. Analyzes material costs, labor, and regulatory compliance to create accurate project bids.

Minimum Education: Associate degree or equivalent experience